

Who Should Be in the Loop?

Allocating Expert Review for Qualitative Coding with Language Models

Anonymous ACL submission

Abstract

Large language models can assist qualitative coding, but specialist review does not scale easily. Existing studies usually report human feedback without asking which kind of expertise is needed for a particular judgment. We present **WarrantRoute**, a source grounded evaluation and routing method for qualitative interpretations produced with language models. Researchers without specialist expertise, qualitative methods experts, and domain experts review natural outputs and controlled variants containing unsupported evidence, concentrated source or speaker representation, missing counterevidence, lost context, or overly broad claims. Using [N PACKETS], [N RATERS], and a held out [DOMAIN AND MODEL], WarrantRoute [MAIN ROUTING RESULT]. Feedback selected by the method [REPAIR RESULT] while requiring [EXPERT TIME RESULT]. The results show [ONE SENTENCE INTERPRETATION]. This study provides evidence about when specialized review changes source based judgments and how a research team can allocate limited expert time while retaining less frequent and negative cases.

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1 Introduction

Large language models (LLMs) can suggest codes, organize codebooks, and apply qualitative categories across large collections of text (Dai et al., 2023; Wang et al., 2026; Matveyenko et al., 2026). Researchers often review these outputs, edit suggestions, explain disagreements, and guide later analysis. Most evaluations still treat their judgments as one general source of human feedback, even when the task calls for different kinds of knowledge.

A qualitative methods expert may recognize that a topic label has been presented as an interpretive theme. A domain expert may notice that an analysis has lost a cultural distinction or omitted a clinically

important exception. A researcher without either specialization may still identify an unsupported quotation or an overly broad claim. A single average score conceals these differences and provides little guidance about how to use limited expert time.

Evaluations of LLM supported clinical thematic analysis remain fragmented across expert review and automatic similarity measures (Yi et al., 2025a). The usual measures do not resolve this problem. More than one code space may be defensible (Chen et al., 2026), agreement between humans and LLMs may not predict blind expert preference (Liu et al., 2026), and model confidence can be poorly calibrated (Li et al., 2026). Audits have also found missing participant evidence, summaries presented as quotations, and failures that depend on local context (Parkington et al., 2025; Bang et al., 2026; Subbiah et al., 2026). Provenance identifies the source of a passage. It does not show that the passage supports the interpretation or represents the range of views in the corpus.

Our experiment treats review as an *allocation problem*: who should examine a qualitative interpretation and its supporting evidence? **WarrantRoute** collects judgments from researchers without specialist expertise, qualitative methods experts, and domain experts. The study includes natural model outputs and controlled variants with known defects. A simple policy sends each item to no specialist, a qualitative methods expert, a domain expert, or both. The policy uses information available before specialist review, and all comparisons receive the same amount of expert time. We fix the policy before testing it on a new domain and model family. Figure 1 summarizes the framework.

This paper makes three contributions:

1. a source grounded benchmark that isolates five validity problems without assuming a single correct thematic interpretation;
2. a human evaluation that adapts evidential cred-

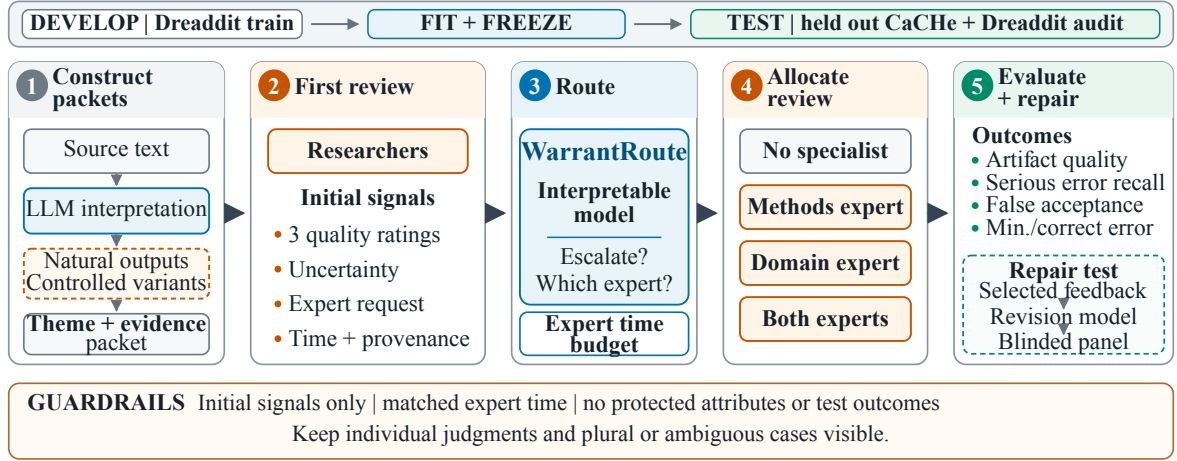


Figure 1: Overview of WarrantRoute. An interpretable policy routes source grounded theme and evidence packets to researchers alone, a qualitative methods expert, a domain expert, or both within a fixed time budget. We evaluate routing on CaChe and Dreddit and test whether expert feedback repairs errors without creating new ones.

ibility and plural-reference coverage from prior work and adds study-specific, source-grounded operationalizations of voice-and-boundary preservation and scope calibration; and

3. a study held out domain test that compares several allocation rules at equal cost and measures whether the selected feedback repairs a defect without creating a new one.

2 Related Work

Human feedback in qualitative analysis.

Human-in-the-loop systems let analysts revise codes and rationales (Dai et al., 2023), pair-code and discuss disagreements (Gao et al., 2024), convert repeated feedback into persistent instructions (Wang et al., 2026), and retain positionality and revision histories (Ye et al., 2026). Multi-agent systems extend this workflow: Auto-TA automates source-to-theme generation (Yi et al., 2025c), SFT-TA uses supervised fine-tuned agents (Yi et al., 2025b), and TAMA incorporates cardiac expert guidance (Xu et al., 2026). These studies establish scalable collaboration, but their evaluations emphasize alignment, usability, control, or aggregate output quality rather than reviewer specialization.

Review allocation and evaluator dependence.

Selective-review methods route items using confidence and ensemble diversity (Zhao and Liu, 2025) or rationale disagreement (Tajik et al., 2026), although confidence can be miscalibrated or anchor-

sensitive (Li et al., 2026). Evaluator identity also changes conclusions: blind expert verification can diverge from agreement metrics (Liu et al., 2026), and specialists who read source material can differ from nonspecialists who see isolated themes (Moskalewicz et al., 2026). Adjacent biomedical work evaluates generated protocol pseudocode against a generated reference (Yi et al., 2024), but not the source warrant of qualitative interpretations. Prior work therefore studies allocation signals and evaluator perspective separately, not which expert role a source-linked packet requires under matched expert time.

Source grounding and artifact quality.

Provenance systems preserve links among quotations, codes, themes, and revisions (Ye et al., 2026; Yi et al., 2026), yet models can omit human-identified excerpts, substitute summaries for quotations, or produce paraphrased, truncated, and hallucinated evidence (Parkington et al., 2025; Bang et al., 2026). Computational thematic-analysis evaluations have operationalized credibility through human acceptance and semantic alignment and, later, binary LLM judgments (Qiao et al., 2025; Wang et al., 2026); OpenCodes instead measures coverage relative to a pooled code space (Chen et al., 2026). These constructs are complementary: cited evidence may support a claim that omits dissent, while broad coverage does not ensure appropriate scope. WarrantRoute therefore adapts evidential credibility and plural-reference coverage, adds packet-level operationalizations of voice-and-boundary preservation and scope calibration, and

reports all four separately. The open comparison is whether methods and domain expertise detect complementary source-grounded failures, and whether expertise-aware routing improves detection over expertise-agnostic allocation at the same cost.

3 Task and Study Design

3.1 Evidence packets and controlled variants

The unit of analysis is a **theme and evidence packet**. Each packet contains a qualitative claim, a short explanation, exact excerpts from the privacy reviewed analysis copy with pseudonymous speaker or document identifiers and local context, and a summary of source coverage. Raters judge whether the evidence supports the claim and preserves important differences, counterevidence, and participant voice. The task is a bounded source-warrant and representation assessment. It does not assess a complete reflexive thematic analysis or assume that one thematic account is universally correct (Braun and Clarke, 2021a,b).

We sample natural LLM outputs and construct variants vetted by experts. Each variant changes one preregistered property:

1. **unsupported evidence**: a topically related excerpt does not warrant the claim;
2. **source concentration**: wording implies breadth while evidence comes disproportionately from one participant or document;
3. **counterevidence loss**: a rare, negative, or boundary case is removed;
4. **contextual flattening**: a consequential cultural or domain specific qualification is erased; or
5. **unsupported abstraction**: a local observation is generalized beyond its evidence.

An independent construction panel checks the original packet, the intended defect, and the absence of a second manipulation. Pilot items do not enter the main analysis, and each rater sees at most one version of a base packet.

3.2 Artifact-quality measures

We preregister four artifact-quality measures, separately from the routing and repair outcomes below. Two adapt measures used in prior computational thematic analysis. **Evidential credibility**

asks whether the core interpretive meaning is directly supported by the cited evidence in local context. Unlike prior binary LLM judgments (Qiao et al., 2025; Wang et al., 2026), our primary measure is a blinded human rating. **Plural-reference coverage** is the percentage of concepts in a frozen reference space pooled from multiple independent analysts that are recovered by at least one generated code or theme (Chen et al., 2026). The pool preserves alternative concepts rather than designating one analyst’s themes as truth, and blinded experts adjudicate many-to-many conceptual matches.

We add two study-specific, packet-level operationalizations. **Voice-and-boundary preservation** assesses whether consequential differences, dissent, counterevidence, minority or boundary cases, and contextual qualifications survive synthesis. **Scope calibration** assesses whether claim breadth, strength, polarity, and causal language are proportionate to the supplied evidence. Raters score these two measures and evidential credibility from 1 to 5. We retain the ordinal scores and report ratings of 4 or 5 as descriptively adequate; “cannot judge” is reported separately. Plural-reference coverage is calculated only on complete natural-output inventories, separately for codes and themes, and is not a 1–5 rating. We interpret the four measures only after deterministic quotation, attribution, assignment, and provenance checks. Because coverage is recall-like, we report generated-concept volume and human-audited unsupported or redundant concepts so that verbosity cannot masquerade as quality. We do not combine the four measures into a composite. Full anchors and matching rules appear in Appendix B.4.

3.3 Datasets and leakage controls

The project maintains four candidate corpus lanes that differ in domain, interaction structure, and provenance. Dreddit’s official training split supplies development data, its official test split is a secondary in domain audit, and the 11 CaChE focus groups form the held out cross domain confirmatory evaluation. CaSiNo and the AMI scenario meeting subset replace KODIS and CANDOR as the planned dialogue settings, covering task oriented dyadic negotiation and elicited multiparty meetings. Their exact source snapshots have been acquired and integrity checked, but neither replacement lane yet has adapter validated records, an approved model or rater pathway, or a frozen experimental role or split. Here, “held out” describes

our procedure, not model training provenance.

Table 1 reports each lane at its current preparation stage. Dreddit and CaChe counts come from restricted processed copies, whereas CaSiNo and AMI counts are verified source inventories. The source only rows are not common experimental denominators and are not sampling eligible. Definitions, provenance, access conditions, and calculations appear in Appendix B.1.

Only Dreddit training data inform prompts, packet rules, controlled variants, routing features, and thresholds. We freeze these choices before test generation. We preserve complete source groups, quarantine exact training duplicates that overlap Dreddit test text, and cluster CaChe analyses by focus group and transcript local speaker. CaSiNo and AMI do not inform these choices while their experimental roles remain unassigned. Packets use the same context limits across adopted evaluation datasets. We retain prompts, decoding settings, run identifiers, source spans, and revisions. Protected text is processed only under approved terms. The complete leakage and generation protocol appears in Appendix B.1.

3.4 Raters and annotation

We recruit three groups. **Qualitative methods experts** have demonstrated experience conducting qualitative analysis. **Domain experts** have research, professional, or relevant lived experience in the source domain. **Researchers without specialist expertise** can apply the study rubric but do not meet either expert definition for the assigned item. Dual expertise is recorded as a separate status. Item construction, initial rating, policy development, and final adjudication are handled by different people wherever staffing permits.

All raters complete the same tutorial and review balanced subsets in a randomized incomplete block design. For each packet they rate evidential credibility, voice-and-boundary preservation, scope calibration, and confidence on scales from 1 to 5. The common quality anchors are 1 (not adequate), 2 (major problems), 3 (partially adequate; material revision needed), 4 (mostly adequate; minor issues only), and 5 (fully adequate); confidence uses separate uncertainty anchors. They also choose accept, revise, reject, or escalate; indicate whether they cannot judge the item; provide a short rationale and, if needed, request a particular kind of expertise. The system records review time. “Cannot judge” is a separate response category, with no conversion to a

midpoint or a quality failure. Qualitative methods experts and domain experts submit independent ratings before adjudication. When their disagreement reflects defensible interpretations, the item remains plural or ambiguous.

We choose the final sample size after a separate pilot by simulating the crossed structure of raters and packets. The preregistration states the number of unique raters, base packets, ratings per group, exclusions, and the smallest effect or noninferiority margin that the design can support.

4 Routing by Expertise

The routing method uses information available before specialist review: researchers’ ratings of evidential credibility, voice-and-boundary preservation, and scope calibration, plus disagreement, confidence, abstention, the requested kind of expertise, response time, and nonsensitive provenance measures such as excerpt count, source concentration, and quotation exactness. Plural-reference coverage is evaluation-only because it relies on the frozen analyst reference space. The router cannot use manipulation labels, expert scores from the test set, protected participant attributes, or test outcomes.

We fit an interpretable regularized model that chooses among no escalation, review by a qualitative methods expert, review by a domain expert, or review by both. We select its thresholds on the development data to maximize recall of serious errors within a fixed budget measured in expert minutes. We then compare five conditions:

1. **researchers only**: no specialist review;
2. **both expert groups**: a qualitative methods expert and a domain expert review every item;
3. **random**: route the same budget at random;
4. **uncertainty only**: route using confidence, disagreement, and abstention; and
5. **WarrantRoute**: predict both whether escalation is needed and which role is relevant.

Every condition uses the same observed expert time. The confirmatory outcomes on the CaChe cross domain evaluation are recall of serious errors, false acceptance, and expert minutes per correctly escalated error; we report the same outcomes on the Dreddit in domain audit as secondary results. These policy-level outcomes are distinct from the

Dataset	Domain and context	Language	Sources	Text units	Text count	Status
Dreaddit (Turcan and McKeown, 2019)	Social media stress; short texts	English	2,929 posts	3,553 source; 3,532 eligible	302,509 eligible words	Processed; review pending
CaChE (Mehta, 2024)	Sexual and reproductive health; focus groups	Translated English	11 groups	3,944 retained; 3,076 eligible	98,735 eligible words	Processed; review pending
CaSiNo (Chawla et al., 2021)	Campsite negotiation; dyadic text dialogue	English	1,030 dialogues	11,919 ordinary messages	228,675 source words	Source only
AMI scenario (Carletta et al., 2006)	Product design; four party meetings	Spoken English	138 meetings	69,258 source segments	793,764 source <w> elements	Source only

Table 1: Dataset inventory at the current preparation stage. Dreaddit and CaChE eligible counts apply deterministic content rules but remain pending manual privacy review and project authorization. CaSiNo source counts use ordinary messages and whitespace delimited words and exclude 2,378 structured negotiation events. AMI counts upstream speech segment and orthographic word XML elements. Source only rows are neither processed nor sampling eligible, and the differing unit types are not common analysis denominators.

four artifact-quality measures. A secondary experiment uses one fixed revision model and compares no feedback, feedback from researchers, feedback selected by WarrantRoute, and feedback from both expert groups. A separate blinded panel judges whether the revision fixes the target defect and whether it creates a new unsupported claim, omission, or distortion.

5 Evaluation and Analysis

The analysis addresses three questions. First, for which errors do qualitative methods experts or domain experts differ materially from researchers without specialist expertise? Second, can the initial ratings identify those cases within a fixed budget of expert time? Third, does the selected feedback repair defects in a new domain and model family?

For the first question, separate cumulative link mixed effects models analyze evidential credibility, voice-and-boundary preservation, and scope calibration; logistic mixed effects models analyze decisions and defect detection. Models include packet version, evaluator group, error family, domain, and their preregistered interactions, with random effects for rater and base packet. Plural-reference coverage is estimated separately at the complete-run level, not from sampled packets or controlled twins. For the second question, we compare policies on the cross domain evaluation with cluster bootstrap intervals and a preregistered noninferiority contrast against review by both expert groups. For the third question, a logistic mixed model relates successful repair to feedback condition and error family. Dreaddit results are secondary; the frozen CaChE analysis is confirmatory.

We report effect sizes and uncertainty intervals. Ordinal Krippendorff’s α , Gwet’s AC2, raw agreement, and confidence calibration describe measurement behavior; they do not establish validity on their own (McDonald et al., 2019). We freeze confirmatory contrasts, multiplicity, and the CaChE packet manifest before generating model outputs for that corpus. Development and audit results, comparisons of natural and controlled items, and other robustness checks appear in Appendix D. Table 7 reports reliability, and Table 8 reports transfer and routing feature ablations. Confidence and router calibration are reported separately in the appendix.

6 Results

This section separates the completed synthetic measurement-pipeline qualification from the pre-registered real-corpus reporting shell. The qualification used four fictional templates from an earlier corpus design, including KODIS-shaped and CANDOR-shaped proxies. It tests whether the four measures can be calculated reproducibly; it provides no evidence about Dreaddit, CaChE, CaSiNo, AMI, or any other real corpus. Replace every later blank from a single frozen analysis export and reconcile all denominators before writing the final abstract. Report the CaChE confirmatory results before the Dreaddit audit and development performance.

6.1 Synthetic measurement-pipeline qualification

We applied the paper-aligned measures to 60 outputs generated from entirely fictional packets: five

recent model variants, four corpus-shaped packets, and three independent runs per model and packet. Three blinded model judges independently scored evidential credibility, voice-and-boundary preservation, and scope calibration using the manuscript anchors. For this engineering pilot, the benchmark’s frozen designed constructs instantiate the reference space; each judge matched every reference separately to generated codes and themes, and a two-of-three panel rule determined recovery. Table 2 reports all four measures.

The pilot exposes a ceiling effect that would be hidden by reporting adequacy alone. GPT-5.6 Sol, GPT-5.5, and GPT-5.6 Terra received 100% adequacy on all three ordinal measures, while their deterministic integrity pass rates were 10/12, 9/12, and 5/12. Code coverage ranged only from 93.7% to 95.2%, and theme coverage from 85.7% to 93.7%. We therefore retain the four constructs as the source-grounded artifact profile but do not use them alone for model selection: integrity failures are exclusionary, full ordinal distributions and controlled-defect sensitivity are reported, and coverage is accompanied by output-volume and unsupported-concept diagnostics. Under that gate-first rule, GPT-5.6 Sol remains the remediation candidate rather than a frozen paper model.

6.2 Study accounting

We recruited [N TOTAL RATERS] raters and analyzed [N RATINGS] judgments over [N PACKETS] packets. [ONE SENTENCE ON EXCLUSIONS, BALANCE, AND THE TEST SET]. Table 3 summarizes study counts by experimental role.

6.3 Artifact-quality profile

On the held out CaChE evaluation, [CREDIBILITY ADEQUACY RATE], [VOICE-AND-BOUNDARY ADEQUACY RATE], and [SCOPE-CALIBRATION ADEQUACY RATE] of evaluable judgments were adequate. Plural-reference coverage was [CODE COVERAGE] for codes and [THEME COVERAGE] for themes. We report these measures with the integrity prerequisite, denominators, and uncertainty, and provide full ordinal distributions, cannot-judge rates, matching uncertainty, unmatched reference concepts, and generated-concept volume in Appendix B.4.

6.4 Where expertise changes judgments

[SUMMARIZE THE INTERACTION BETWEEN EVALUATOR GROUP AND FAILURE FAMILY, INCLUDING UNCERTAINTY]. Table 4 reports predicted probabilities of detecting a defect. [STATE WHETHER THE TWO EXPERT GROUPS PROVIDED COMPLEMENTARY, REDUNDANT, OR INCONCLUSIVE JUDGMENTS].

6.5 Routing under matched expert time

At the preregistered budget of [B EXPERT MINUTES], WarrantRoute [PRIMARY TEST RESULT]. Compared with uncertainty routing, it [EFFECT WITH 95% CONFIDENCE INTERVAL]. Compared with review by both expert groups, it [NONINFERIORITY RESULT]. Table 5 compares the five routing policies under the matched expert time budget. Report results by failure family if an average gain conceals misses on negative cases.

6.6 Does routed feedback repair defects?

WarrantRoute feedback repaired [RATE AND CONFIDENCE INTERVAL] of packets without introducing a new error. The corresponding rate was [RESEARCHER FEEDBACK RATE] for feedback from researchers and [BOTH EXPERT GROUPS RATE] for feedback from both expert groups. Table 6 reports repair outcomes, new errors, credibility changes, and expert time. [REPORT PERFORMANCE RELATIVE TO EXPERT MINUTES AND THE MOST IMPORTANT FAILURE CASE].

7 Discussion

[STATE WHICH EVALUATOR GROUP ADDED VALUE FOR WHICH FAILURE TYPES, WITH EFFECTS]. [STATE WHETHER THE TWO EXPERT GROUPS PROVIDED COMPLEMENTARY JUDGMENTS]. At the observed review times, this produced [PRACTICAL COST AND EXPERT-REVIEW COVERAGE CONSEQUENCE]. These conclusions apply to the evidence review task, failure families, and evaluator groups studied here.

On the new domain and model family, [STATE THE TRANSFER RESULT AND ITS INTERPRETATION]. [STATE WHAT THE RESULT IMPLIES FOR ROUTING BEYOND

Model	Integrity pass	Evidential adequacy	Voice/boundary adequacy	Scope adequacy	Code coverage	Theme coverage	Serious error
GPT-5.6 Sol	10/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	85.7 [78.5, 93.4]	0.0 [0.0, 0.0]
GPT-5.4	9/12	100.0 [100.0, 100.0]	97.2 [91.7, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	93.7 [89.1, 98.4]	5.6 [0.0, 13.9]
GPT-5.5	9/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	92.1 [87.3, 96.8]	0.0 [0.0, 0.0]
GPT-5.6 Luna	8/12	80.6 [69.4, 91.7]	80.6 [69.4, 91.7]	88.9 [80.6, 97.2]	95.2 [90.9, 100.0]	85.7 [79.7, 91.9]	8.3 [0.0, 16.7]
GPT-5.6 Terra	5/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	95.2 [90.9, 100.0]	88.9 [84.1, 93.7]	0.0 [0.0, 0.0]

Table 2: Paper-aligned measurement pilot on entirely fictional packets. Ordinal columns report the percentage of 36 blinded model-judge ratings per model scored 4 or 5. Code and theme coverage report panel-majority recovery across 63 frozen reference opportunities per level and model. Brackets are 95% packet-by-run cluster-bootstrap intervals. Integrity is deterministic. All three judges and candidates are from one provider; these are measurement-pipeline results, not real-corpus or human-evaluation results.

Quantity	Dev.	In domain	Cross domain
Source participants/documents	—	—	—
Base packets / controlled twins	—	—	—
Researchers without specialist expertise	—	—	—
Qualitative methods experts	—	—	—
Domain experts	—	—	—
Independent ratings	—	—	—
Median review minutes	—	—	—

Table 3: Study counts by experimental role. Raters and packets are unique counts.

GENERIC UNCERTAINTY]. A useful review queue should preserve the reason for escalation and allow a reviewer to request another form of expertise. Persistent disagreement remains visible for deliberation.

8 Limitations

Our estimates reflect the recruited reviewers, whose judgments may reproduce disciplinary, cultural, or institutional assumptions. Controlled variants isolate individual failure mechanisms, but naturally occurring errors may be subtler or combine several defects. Repeated judgments from the same people increase precision without increasing the diversity of expertise. Routing can also create false reassurance around items that are not escalated, so we retain a random audit and report false acceptance for each failure family. The task covers bounded evidence packets. A complete qualitative analysis also requires familiarization, memoing, reflexivity, and accountable interpretation. Generalization is limited to English Reddit stress segments and English translations of focus groups from western Kenya, as well as **[MODEL FAMILIES AND RECRUITMENT SAMPLE]**. If the replacement dialogue lanes are later adopted, CaSiNo cannot support cross-dialogue participant counts or participant-disjoint splits from its public release, while AMI scenario meetings are elicited role play and transcript-only processing omits prosody, tim-

ing, overlap, and multimodal evidence.

9 Ethics Statement

Real text model processing and rater exposure begin only after a documented institutional determination and all source, model provider, rater access, retention, and release gates are satisfied; source collection approvals do not transfer. We treat all corpora as potentially identifying. Analysis copies are minimized and restricted, protected text goes only to approved services, and every excerpt shown to raters or released receives privacy review. Raters participate voluntarily outside supervisory or evaluative relationships, with identifiers stored separately. Public artifacts contain only aggregate, synthetic, or cleared material. Routing supports research review, not clinical or participant decisions. Details appear in Appendix C.

10 Conclusion

WarrantRoute measures how reviewer expertise affects the detection and repair of unsupported qualitative interpretations under a fixed review budget. **[STATE THE PRIMARY RESULT AND ITS PRACTICAL IMPLICATION]**.

Failure family	Researchers	Methods experts	Domain experts	Largest contrast
Unsupported evidence	_____	_____	_____	_____
Source concentration	_____	_____	_____	_____
Counterevidence loss	_____	_____	_____	_____
Contextual flattening	_____	_____	_____	_____
Unsupported abstraction	_____	_____	_____	_____

Table 4: Predicted probability of detecting each failure, with 95% confidence intervals, by evaluator group.

Policy	Expert minutes	Serious error recall	False acceptance	Difference from both experts	Errors per expert hour
Researchers only	0	_____	_____	_____	Not applicable
Random	_____	_____	_____	_____	_____
Uncertainty routing	_____	_____	_____	_____	_____
WarrantRoute	_____	_____	_____	_____	_____
Both expert groups	_____	_____	_____	Reference	_____

Table 5: Routing performance on the CaCHe confirmatory evaluation at matched budgets measured in expert minutes. Intervals use a bootstrap clustered by base packet.

Feedback	Repair	New error	Credibility change	Expert minutes
None	_____	_____	_____	0
Researchers	_____	_____	_____	0
WarrantRoute	_____	_____	_____	_____
Both expert groups	_____	_____	_____	_____

Table 6: Results of blinded revision. A repair is successful only if it fixes the targeted defect without introducing an unsupported claim.

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	A Appendix Roadmap	768
	The appendix is intentionally skeletal until the protocol is fixed. Complete the subsections below without moving secondary detail back into the eight page main paper.	769
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	B Full Study Protocol	773
	B.1 Corpus provenance, preprocessing, and leakage controls	774
		775
	The project documents four candidate corpus lanes. Dreddit’s official train partition is development data, its official test partition is a secondary in domain audit, and all 11 CaCHe focus groups are the held out cross domain confirmatory evaluation.	776
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CaSiNo and the AMI scenario meeting subset replace KODIS and CANDOR as the two dialogue corpus candidates. Exact source snapshots for both replacements have been acquired and integrity checked, but their experimental roles remain unassigned. Neither has an adapter validated output, an approved model or rater pathway, a completed contextual privacy review, or a frozen project split.

Dreaddit is a publicly downloadable English Reddit corpus for stress detection. Its released labeled subset contains 3,553 human annotated segments from 2,929 posts across ten subreddits in five domains (Turcan and McKeown, 2019). The archive provides official train and test partitions and post identifiers, but no author identifier. It therefore supports segment to post provenance and document concentration checks, not participant counts or participant level claims. The author archive contains 2,838 train segments from 2,343 posts and 715 test segments from 586 posts. It contains no README or dataset license. We retain text, the stress label, subreddit, and sentence range in the restricted analysis copy. We omit original post identifiers, timestamps, engagement measures, the internally inconsistent confidence field, and unused lexical features. Project specific identifiers preserve post grouping without exposing the released post identifier. The official splits have no post identifier overlap, but three exact texts occur in both partitions under different identifiers and conflicting labels. We retain those rows for audit but make the three training copies and 18 later within train exact duplicates ineligible for sampling, leaving 2,817 development segments and 715 audit segments.

The CaChE deposit contains 11 participant numbered and selectively redacted focus group transcripts about how COVID-19 affected schooling, sexual behaviour, and sexual and reproductive health in western Kenya (Mehta, 2024). The source study reports six groups with 54 adolescent girls and young women and five with 53 community men. Sessions used English, Swahili, Dholuo, or a mixture and were translated into English (Awiti et al., 2024). The deposit is version 1, DOI 10.25417/uic.26495884.v1, and CC BY 4.0. A fixed checksum allowlist verifies all 11 source files. Processing retains 3,575 participant turns and 369 collective turns; 3,076 participant turns meet the earlier three word eligibility rule, but short and collective turns are no longer deleted and remain available as context. Each record has a stable focus group and turn identifier, a transcript local speaker

identifier, source line spans, source hashes, translation status, the preceding moderator question, and prior turn references. Normalization yields 106 observed transcript local participant keys, one fewer than the 107 participants reported by the source paper; we do not claim complete participant coverage until this is reconciled. Conflicting source dates are recorded but excluded from model and router features.

The acquired CaSiNo source is the official CC BY 4.0 repository snapshot pinned to commit 2f6ed4a. It contains 1,030 dyadic campsite negotiation dialogues, 11,919 ordinary messages, 2,378 structured negotiation events, and 228,675 whitespace delimited words in the ordinary messages (Chawla et al., 2021). The source paper reports 846 participants, but the public JSON provides only two dialogue local role tokens and no stable cross-dialogue participant token. We therefore reject the supplied 900/30/100 random split for WarrantRoute, do not reconstruct linkage from participant profiles or text, and leave the corpus unassigned unless trustworthy provider linkage becomes available or the entire corpus is preregistered in one unsplit role after the governance gates are satisfied. Any future adapter will retain ordinary dialogue text and turn lineage only. It will exclude structured events, annotations, demographics, personality measures, preferences, free text reasons, negotiation outcomes, and raw identifiers.

The acquired AMI manual annotation artifact contains a selectively extracted source inventory for 138 elicited scenario meetings in 35 meeting series. The four person meetings are English product design discussions. The selected files contain 793,764 orthographic word elements, 69,258 speech segment elements, and mappings for 140 global source speaker identifiers. These form 35 meeting and speaker connected components, 33 with four meetings and two with three meetings (Carletta et al., 2006). No audio, video, signals, slides, questionnaires, demographics, participant file, or nontranscript annotation category was extracted. The official filename labels the manual annotations as version 1.6.2, while the embedded README says release 1.7; we bind the source by its exact URL, filename, size, and checksum rather than asserting that those labels are equivalent. The corpus is not sampling eligible. A future transcript adapter must discard timing and raw source identifiers, and all 35 meeting and speaker components must remain indivisible in any adopted split.

For Table 1, a source is one Reddit post, focus group, CaSiNo dialogue, or AMI meeting. Dreddit source units are released labeled segments, and its eligible units exclude three training copies that overlap test text plus 18 later within-train duplicates. CaChe retained units include participant and collective turns, while its eligible count applies the earlier three word participant-turn rule. CaSiNo units are ordinary source messages and exclude structured events. AMI units are upstream speech segment elements. CaSiNo and AMI counts are source inventories, not processed turns, sampling eligible units, or common analysis denominators.

Dreddit’s official train partition is the only corpus used to develop prompts, packet rules, controlled variants, routing features, and thresholds. After these choices are frozen, its official test partition provides an in domain audit. All 11 CaChe focus groups are held out from WarrantRoute development and form the cross domain confirmatory evaluation. “Held out” refers to our experimental procedure, not model training provenance. The deposit is public and published themes are known, so we do not call it unopened or model unseen and do not use the published themes as scoring gold.

The power analysis and expert time budget determine a common number of base packets for the Dreddit audit and CaChe evaluation. Sampling preserves post clusters in Dreddit and focus group and transcript local speaker clusters in CaChe. Both corpora use identical packet and context limits. Dreddit labels and published CaChe themes may stratify audits but are not treated as thematic ground truth or router features. CaSiNo and AMI enter no generation, sampling, or model selection procedure unless a later preregistration assigns them an experimental role after corpus specific validation and governance approval.

We generate packets using [MODEL VERSIONS] and [GENERATION PROCEDURE]. We retain prompts, decoding settings, context order, run identifiers, source spans, and human revisions. We do not select test outputs after inspecting their quality. We use text or transcripts only, process them under the approved governance arrangement, and do not send protected material to an external model unless the access terms and our institution permit it. Material released publicly is synthetic or cleared for release.

[CURRENT INSTITUTIONAL DETERMINATION; SOURCE, PLATFORM, MODEL PROCESSING, AND RATER ACCESS AU-

THORIZATIONS FOR DREADDIT, CACHE, CASINO, AND AMI; CASINO LINKAGE RESOLUTION; AMI COMPONENT SPLIT; RATER EXCERPT RULES; FINAL PACKET COUNTS; MODEL SNAPSHOTS; PROMPTS; DECODING SETTINGS; RUN COUNTS; AND FROZEN MANIFEST HASHES].

B.2 Role definitions and recruitment

[ELIGIBILITY FOR RESEARCHERS WITHOUT SPECIALIST EXPERTISE, QUALITATIVE METHODS EXPERTS, AND DOMAIN EXPERTS; HANDLING OF DUAL EXPERTISE; RECRUITMENT; COMPENSATION; POSITIONALITY; EXCLUSIONS; AND INDEPENDENCE SAFEGUARDS].

B.3 Complete annotation instrument

The three packet-level quality questions are: (1) *How fully does the cited evidence support the core interpretive meaning in context?* (evidential credibility); (2) *How well does the packet preserve consequential differences, dissent, counterevidence, minority or boundary cases, and contextual qualifications?* (voice-and-boundary preservation); and (3) *How well do the strength and breadth of the claim, including causal or group-level language, match the supplied evidence?* (scope calibration).

For each question, 1 means not adequate, 2 major problems, 3 partially adequate and requiring material revision, 4 mostly adequate with only minor issues, and 5 fully adequate. The tutorial supplies construct-specific examples at each anchor. For rating r_{ijm} from rater i on packet j , we analyze the full ordinal response; the descriptive adequacy rate for measure m is the proportion of evaluable rater–packet judgments scored 4 or 5:

Evidential credibility. A 1 indicates absent, unverifiable, misattributed, or contradictory evidence; a 3 indicates that the core meaning is partly supported but a material link or qualifier lacks evidence; and a 5 indicates that every material part of the core interpretation is directly warranted by sufficient contextual evidence.

Voice-and-boundary preservation. A 1 indicates that a consequential voice, counterexample, or contextual distinction is erased or reversed; a 3 retains the main pattern but underplays a meaningful viewpoint or boundary; and a 5 retains all consequential

variation, dissent, minority cases, and contextual qualifications visible in the packet.

Scope calibration. A 1 indicates that claim breadth, polarity, or causal strength is incompatible with the evidence; a 3 has the broadly appropriate direction but requires a material change in scope or strength; and a 5 fully matches participant-, group-, and corpus-level scope and makes no stronger causal or prevalence claim than the evidence permits.

Scores 2 and 4 represent the corresponding intermediate judgments under the common anchors. Raters select “cannot judge” when the supplied excerpts, local context, or source-coverage summary are insufficient for the specified dimension.

$$A_m = \frac{\sum_{i,j} \mathbb{I}(r_{ijm} \in \{4, 5\})}{\sum_{i,j} \mathbb{I}(r_{ijm} \text{ is evaluable})}. \quad (1)$$

“Cannot judge” is neither a midpoint nor a failure and is reported with its own denominator. Confidence has separate anchors from 1 (very uncertain) to 5 (very certain). Exact-string and source-offset checks are recorded as deterministic diagnostics; raters judge whether verified excerpts warrant the interpretation in context.

[CONSENT TEXT, TUTORIAL EXAMPLES, COMPREHENSION CHECKS, ACCEPT/REVISE/REJECT/ESCALATE WORDING, ABSTENTION RULE, RATIONALE PROMPT, AND TIMING DISCLOSURE].

B.4 Artifact-quality definitions and matching

Plural-reference coverage is calculated only for complete, natural-output code and theme inventories. It is not calculated from sampled packets or controlled twins. Before inspecting test outputs, multiple analysts independently produce code and theme concepts. We freeze their pooled reference space after merging only concepts adjudicated to be equivalent, retaining the originating analyst and source provenance. This pool represents a bounded set of defensible concepts available to the study team, not universal thematic ground truth.

For output level $\ell \in \{\text{code, theme}\}$, let P_ℓ be the frozen pooled concepts and G_ℓ the generated concepts. Independent experts blinded to system condition may make many-to-many conceptual matches, denoted by $p \sim g$. Coverage is

$$C_\ell = \frac{|\{p \in P_\ell : \exists g \in G_\ell, p \sim g\}|}{|P_\ell|}. \quad (2)$$

We report code and theme coverage separately with preregistered uncertainty intervals, list unmatched reference concepts, retain disagreements about matching, and do not count unmatched generated concepts as errors without separate evidential review. To diagnose coverage gained through indiscriminate output, we also report generated code and theme counts, unmatched generated concepts, split and merge patterns, redundancy, and the human-audited unsupported-concept rate. Because the measure relies on the analyst reference space, it is not available to the router. The three ordinal measures and plural-reference coverage remain separate; no omnibus quality score is computed.

For controlled variants, the primary manipulation check maps unsupported evidence to evidential credibility; source concentration, counterevidence loss, and contextual flattening to voice-and-boundary preservation; and unsupported abstraction to scope calibration. We also report cross-construct effects rather than assuming that each defect can affect only one measure. Plural-reference coverage is restricted to natural-output inventories and is not a manipulation check for controlled twins.

B.5 Packet construction and assignment

[NATURAL ITEM SAMPLING, CONSTRUCTION OF CONTROLLED VARIANTS, VETTING, MANIPULATION CHECKS, RANDOMIZATION, INCOMPLETE BLOCK ASSIGNMENT, AND BLINDING].

B.6 Power and confirmatory analysis

[PILOT VARIANCE COMPONENTS, SIMULATION DESIGN, POWER TARGET, NON-INFERIORITY MARGIN, MODEL EQUATIONS, MULTIPLICITY PLAN, FALLBACKS, AND EXCLUSIONS].

B.7 Routing implementation

[FEATURE DEFINITIONS, PREPROCESSING, MODEL, HYPERPARAMETERS, BUDGET SELECTION, THRESHOLDS, CROSS FITTING, FREEZE DATE, AND TEST ACCESS LOG].

C Detailed Ethics and Data Governance

C.1 Source review and access conditions

Approval of an original collection does not authorize this secondary study, model processing, or dis-

closure to raters. Before analysis, we will record our institution’s determination, its scope and date, and the governing source, platform, and access terms for each corpus.

The CaChe source study reports written consent and review by Rush University Medical Center (22011311), Maseno University Ethics Review Committee (MSU/DRPI/MUERC/01021/21), and University of Illinois Chicago (2022-0220), with a favorable opinion from Liverpool School of Tropical Medicine (21-087) (Awiti et al., 2024). The CaSiNo paper reports institutional review, informed consent covering collection and later use, a right to withdraw, and removal of MTurk and HIT identifiers before public release (Chawla et al., 2021). Although that paper calls the released data anonymized, we do not adopt that label: dialogue and rich participant metadata can remain identifying. The AMI site documents European Commission ethics review and signed consent that anticipated possible release to the wider research community; its consent form states that anonymity could not be guaranteed and allowed participants to request removal (AMI Corpus, n.d.). The CC BY 4.0 licenses for CaSiNo and AMI do not grant privacy or personality rights or authorize this project’s model processing, rater disclosure, retention, or release. We did not find an ethics determination, Reddit author consent, privacy procedure, or dataset license in the Dreddit paper or author archive (Turcan and McKeown, 2019); any use therefore remains conditional on a documented institutional and platform basis.

C.2 Project safeguards and rater protections

We do not describe the corpora as anonymous. Dreddit retains searchable verbatim posts. CaChe contains sensitive sexual and reproductive health narratives that may remain identifying despite participant numbers and selective redaction. CaSiNo raw records combine dialogue with demographics, personality measures, preferences, free text reasons, and outcome surveys; those metadata remain excluded from any future working output. AMI source XML contains transcript text, precise timing, and attribute bearing source identifiers; future working outputs must omit timings and raw identifiers. Restricted analysis copies use project pseudonymous identifiers, minimize fields, mask direct identifier patterns, and preserve protected linkage information separately. Pattern masking does not replace contextual privacy review. Pro-

tected text is not sent to a model or service unless its processing, retention, and training terms are approved. [STATE STORAGE LOCATION, ACCESS ROLES, RETENTION PERIOD, DELETION PROCEDURE, APPROVED MODEL PROVIDERS, AND PROVIDER RETENTION OR TRAINING SETTINGS].

Every excerpt shown to raters or considered for release receives a two person privacy and permissions review. Public artifacts contain aggregate results, code, prompts, schemas, synthetic packets, and only text or ratings cleared for release. We do not publish raw hashes of short sensitive excerpts without assessing linkage risk. Raters receive a study specific consent form and privacy warning, may withdraw under the approved procedure, and participate voluntarily outside supervisory, employment, grading, or authorship decisions. Identifiers are stored separately from responses. [STATE RECRUITMENT CHANNELS, ELIGIBILITY, COMPENSATION AND RATE JUSTIFICATION, WITHDRAWAL HANDLING, RATER DEMOGRAPHICS, AND FINAL CONSENT LANGUAGE]. Routing is a research review aid, not a clinical or participant level decision system; random audits and false acceptance reporting address, but do not remove, the risk of false reassurance.

D Supplementary Results

[ADD CONFUSION MATRICES, SELECTIVE-RISK VERSUS EXPERT-REVIEW-COVERAGE CURVES, RESULTS BY FAILURE FAMILY, COMPARISONS OF NATURAL AND CONTROLLED ITEMS, CONFIDENCE AND ROUTER CALIBRATION PLOTS, RATIONALE ANALYSIS, AND ROBUSTNESS CHECKS].

E Reproducibility and Governance Checklist

- ☐ Confirm institutional, source or platform, model processing, and rater access conditions for Dreddit, CaChe, CaSiNo, and AMI; resolve CaSiNo linkage or its unsplit role; validate AMI transcript preprocessing and freeze component isolated splits; then freeze the analytic contract, prompts, model snapshots, and manifests.
- ☐ Obtain ethics determination and finalize voluntary recruitment and withdrawal procedures

Evaluator group and construct	Krippendorff α	Gwet AC2	Raw agreement	Cannot judge	Median time
Researchers: credibility	—	—	—	—	—
Researchers: voice/boundary	—	—	—	—	—
Researchers: scope calibration	—	—	—	—	—
Methods experts: credibility	—	—	—	—	—
Methods experts: voice/boundary	—	—	—	—	—
Methods experts: scope calibration	—	—	—	—	—
Domain experts: credibility	—	—	—	—	—
Domain experts: voice/boundary	—	—	—	—	—
Domain experts: scope calibration	—	—	—	—	—

Table 7: Reliability shell for the three ordinal artifact-quality measures. Report intervals and score prevalence in the final appendix; confidence and router calibration are reported separately.

Routing configuration	Dreaddit dev.	Dreaddit audit	CaChE	Recall change	Expert minutes
Full features	—	—	—	—	—
No disagreement	—	—	—	—	—
No abstention	—	—	—	—	—
No provenance	—	—	—	—	—
Confidence only	—	—	—	—	—

Table 8: Effects of transfer and removal of routing features. Define the reported metric in the final caption.

that avoid coercion.

- ☐ Freeze role criteria, item manifest, the four metric definitions and anchors, plural-reference construction and matching rules, assignment, exclusions, and power analysis.
- ☐ Freeze router features, thresholds, budgets measured in expert minutes, primary outcomes, and analysis code before generating CaChE model outputs.
- ☐ Release only packets cleared by the two person excerpt review, plus hashes, permitted ratings, prompts, code, model and run metadata, and a datasheet.
- ☐ Replace every \fillin, \blankcell, and \blankwide; set \predatadraftfalse; verify the eight page boundary; and update the abstract last.